Annotated Bibliography

The SPRUCE project studies how northern peatlands (wetlands with moss, shrubs, and trees) will respond to warmer temperatures and higher CO levels in the future. Scientists warm the air and soil up to +9 °C and add extra CO inside large enclosures in a bog in northern Minnesota. They have seen that the important Sphagnum moss (the main peat-forming plant) grows much less and becomes drier, partly because it gets less light when shrubs grow bigger and shade it. In 2021, a separate shading experiment used different densities of shade cloth to mimic shrub shading and measured how this affected Sphagnum growth, photosynthesis, nitrogen, and especially tissue water content. The main question is: Does shading by itself make Sphagnum moss dry out a lot, or is drying mostly caused by higher temperatures? Comparing the two experiments helps show whether light (shading) or temperature (warming and drying) is more important for keeping moss wet and helping the peatland store water and carbon in the future.

The 2021 dataset (Norby et al., 2023) includes three CSV files with measurements collected from May to October 2021 in 30 small plots where researchers tested how different levels of shade affected Sphagnum moss. It reports key information such as new growth, tissue water content, nitrogen concentration, and photosynthesis rates. In contrast, the warming + CO dataset (Petro et al., 2023) consists of two CSV files containing data gathered from 2017-2021 inside the main SPRUCE enclosures that were warmed and enriched with elevated CO. It includes the moss's carbon and nitrogen isotope values, rates of nitrogen fixation and methane processing by associated microbes, and some measurements of moss tissue water content, providing longer-term context on how warming and elevated CO influence Sphagnum over multiple years.

Bengtsson, F., Granath, G., Cronberg, N., & Rydin, H. (2020).

Mechanisms behind species-specific water economy responses to water level drawdown in peat mosses. *Annals of Botany, 126*(2), 219–230. https://doi.org/10.1093/aob/mcaa033

Summary: The researchers conducted a controlled water table manipulation experiment with multiple *Sphagnum* species, measuring physiological traits (water content, photosynthesis, growth) to identify mechanisms underlying species-specific drought responses. They discovered that species varied in their water retention capacity and photosynthetic sensitivity to desiccation, with hummock species showing greater tolerance through morphological and physiological adaptations compared to hollow species. By comparing physiological responses across multiple species, this study effectively explains why certain species occupy wet areas while others can survive in drier locations.

Relevance to my work: Understanding how different species handle drought will help me select appropriate species for my study and explain differences in their productivity under varying moisture conditions.

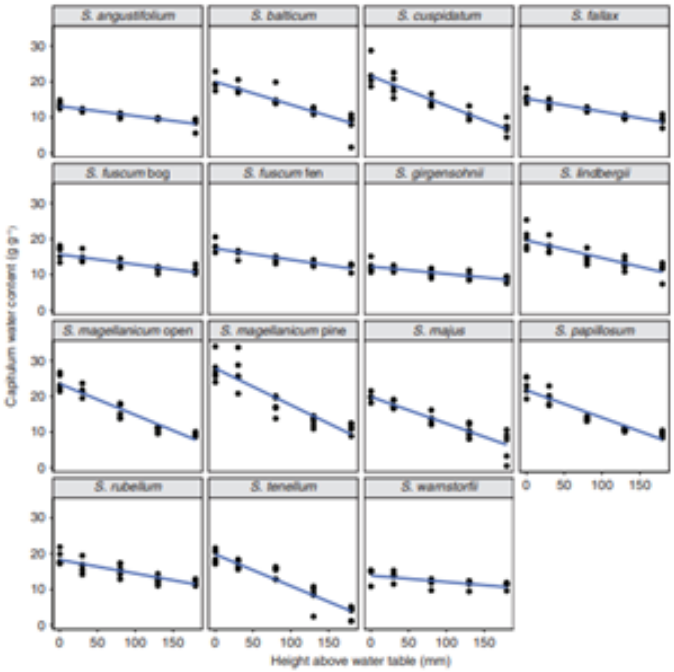


FIG. 2. Capitulum water content measured in samples of 13 *Sphagnum* species as the water table was lowered from 20 to 200 mm below the surface. Lines illustrate the linear relationship between capitulum water content and water level; the slope of the line defines the parameter WC_{slope} (g g^{-1} mm^{-1}) at the species level; n = 4–5 per species. Note that in subsequent statistical analyses, individual values of WC_{slope} were used for each sample. See Fig. 3 for the values of WC_{slope}.

This is a multi-panel scatter plot displays the relationship between water table depth (x-axis) and water content (y-axis) for 15 Sphagnum species. Each panel shows one species with data points and a fitted trend line showing how water content declines as the moss is positioned higher above the water table.

Chong, M., Humphreys, E., & Moore, T. R. (2012). Microclimatic Response to Increasing Shrub Cover and Its Effect on Sphagnum CO2 Exchange in a Bog. *Écoscience (Sainte-Foy), 19*(1), 89–97. https://doi.org/10.2980/19-1-3489

Summary: The researchers manipulated shrub cover in a peatland and measured how shading affected microclimate and Sphagnum carbon exchange using chamber-based gas exchange measurements. They found that increased shrub cover reduced light availability and temperature at the moss surface, leading to decreased net CO uptake by Sphagnum, though this was partially offset by reduced evaporation and maintained moisture. This work is important for understanding vegetation-mediated feedbacks in changing peatlands, with the strength of experimental manipulation but the limitation of relatively short-term measurements that may not capture acclimation responses.

Relevance to my work: This will help me understand how shading from plants affects Sphagnum: less light reduces photosynthesis, but the shade also helps keep the moss moist.

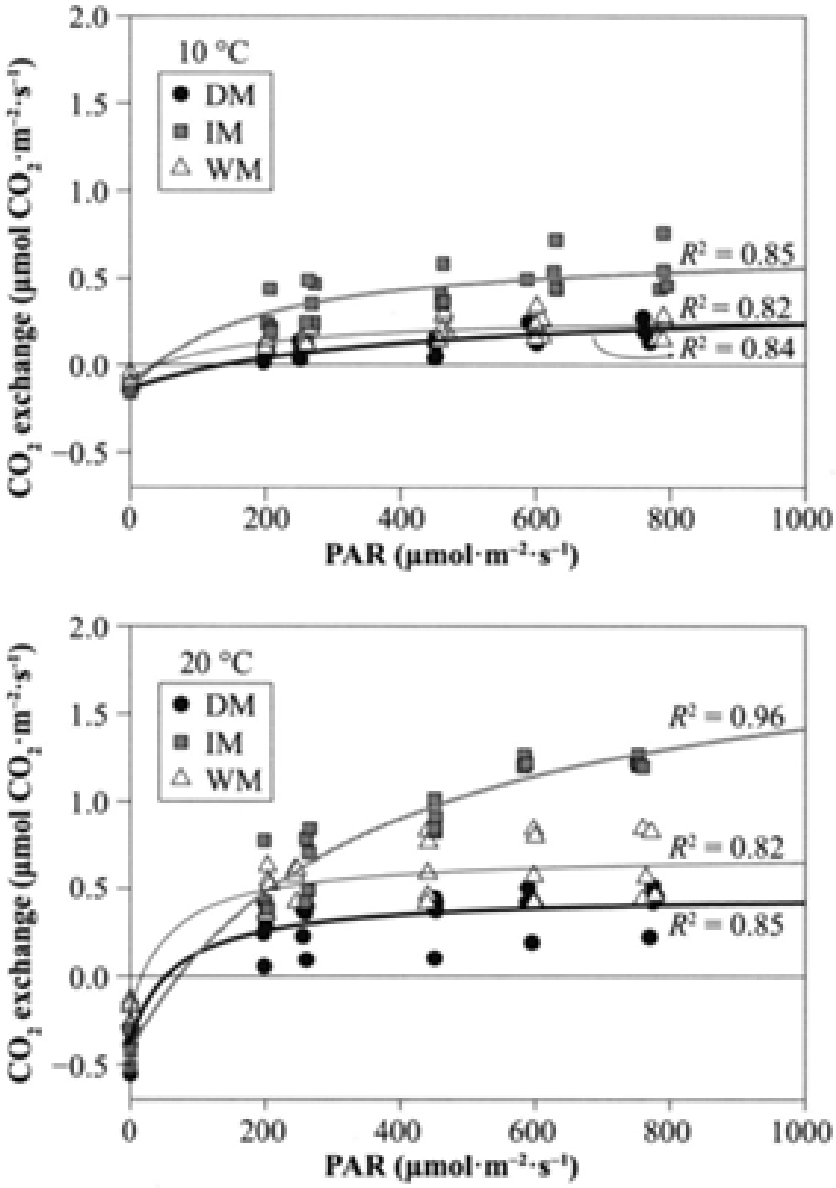


FIGURE 4. The *S. capillifolium* CO_2 exchange response to photosynthetically active radiation (PAR) at 10 and 20 °C and at 3 moisture levels in experiment 2. Gravimetric water contents were dry (DM) < 500%, intermediate (IM) 500 to 2200%, and wet (WM) > 2200%. Lines represent a rectangular hyperbola relationship (Equation [3]). Each data point is 1 CO_2 flux measurement.

This is a light response curve graph, this supports the paper's main conclusion that decreased light availability from increased shrub cover is the primary cause of Sphagnum loss in fertilized plots.

Gauthier, T.-L. J., Elliott, J. B., McCarter, C. P. R., & Price, J. S. (2022). Field-scale compression of Sphagnum moss to improve water retention in a restored bog. *Journal of Hydrology (Amsterdam), 612*, Article 128160. https://doi.org/10.1016/j.jhydrol.2022.128160

Summary: This field-scale restoration experiment tested whether mechanical compression of Sphagnum could improve water retention and establishment success by reducing pore size and capillary rise capacity.

Results demonstrated that compressed Sphagnum maintained higher water content during dry periods and showed improved establishment compared to uncompressed controls, suggesting a practical management technique for restoration. The applied field-scale approach is valuable for restoration practice, though the study could benefit from longer-term monitoring to assess whether compression effects persist and whether there are trade-offs in terms of growth rates or carbon accumulation.

Relevance to my work: This study shows how compressing moss affects its water retention, which could help inform experimental approaches or explain natural differences in moss packing and structure.

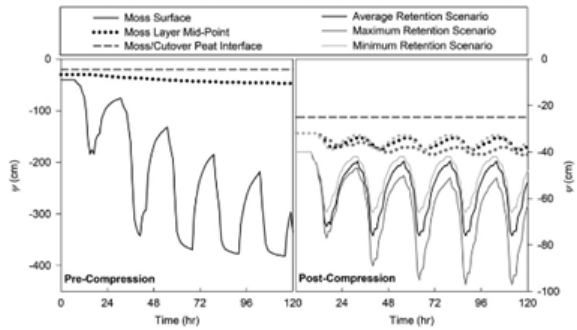


FIGURE 4 Pre-compression and post-compression (avg, min, and max) pressure response to modelled 2.4 mm/day evaporation in HYDRUS-1D. Decrease in pressure (i.e., more negative) indicates increased water stress. Please note the different y-axis scales

This dual-axis line graph shows how water depth and pressure change over time in a peat reservoir during excavation. The solid and dashed lines track different soil layers on the left axis, while the gray lines show pressure scenarios on the right axis. The graph clearly shows that as excavation continues, water stress increases. This is visible through two changes: depths decrease and pressures become more negative.

Goetz, J. D., & Price, J. S. (2015). Role of morphological structure and layering of Sphagnum and Tomenthypnum mosses on moss productivity and evaporation rates. *Canadian Journal of Soil Science*, 95(2), 109–124. https://doi.org/10.1139/CJSS-2014-092

Summary: This laboratory study examined how different moss species' morphological characteristics (capitulum density, layering, porosity) influenced water retention, evaporation rates, and photosynthetic productivity through controlled experiments. Results showed that Sphagnum species with denser capitula and greater water-holding capacity maintained higher productivity under drying conditions compared to less structured mosses, with morphology explaining much of the interspecific variation in drought tolerance. The controlled experimental design is a strength, allowing isolation of morphological effects, though extrapolation to field conditions with additional environmental complexities requires caution.

Relevance to my work: Understanding how different Sphagnum species' physical traits affect water retention is important for choosing the right species to study and for explaining differences in productivity under wet versus dry conditions.

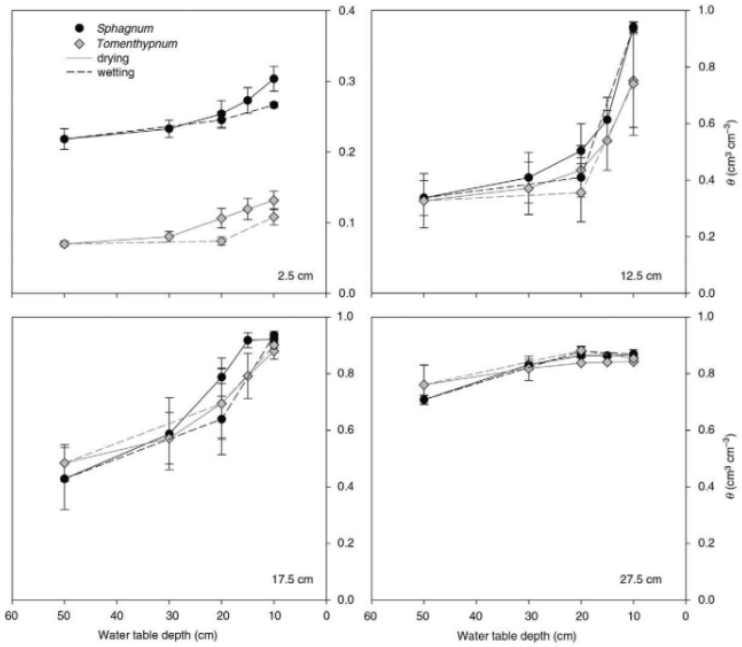


Fig. 6. Changes in mean (±1 standard deviation) θ at 2.5, 12.5, 17.5, and 27.5 cm depth ($n = 3$ monoliths per moss and depth) of *Tomenthypnum* and *Sphagnum* with each subsequent water table change. Water table depths of 10, 15, 20, 30, and 50 cm (solid lines) and rewetting to 20 and 10 cm (dashed lines) were used. Note the different scale used for θ at 2.5 cm than the other depths.

This graph shows how water content changes over time at four depths in Tomenthypnum and Sphagnum moss as the water table is lowered and then raised. Sphagnum holds much more water than Tomenthypnum, especially near the surface (2.5 cm). When the water table is raised from 50 cm back to 20 cm, Sphagnum rehydrates, but Tomenthypnum stays dry.

Keane, B., Shuttleworth, E. L., Evans, M. G., Ritson, J. P., Harris, A., Johnston, A., Alderson, D. M., & Clay, G. D. (2025). Recovery of Sphagnum from drought is controlled by species-specific moisture thresholds. *Scientific Reports*, 15(1), Article 22167. https://doi.org/10.1038/s41598-025-05348-8

Summary: This study monitored Sphagnum recovery following natural drought events across multiple species, identifying critical moisture thresholds below which recovery was impaired or failed for each species. The researchers found that species occupying different microhabitats had distinct moisture thresholds for recovery, with hollow species requiring higher moisture levels for successful recovery compared to hummock species, and that exceeding these thresholds led to prolonged recovery times or mortality. The longitudinal field observations following natural drought events provide high ecological relevance, though the reliance on natural variation limits experimental control and causal inference.

Relevance to my work: Identifying moisture thresholds for each species is important for designing experiments, especially when choosing realistic drought treatments and predicting how well species will recover under different water conditions.

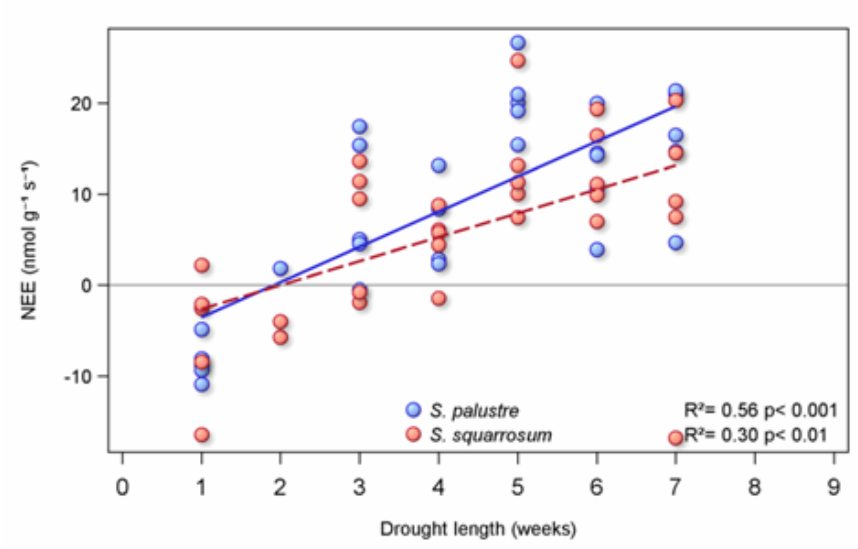


Fig. 3. Relationship between length of drought period and fluxes of CO_2 from two *Sphagnum* species (*S. palustre*, blue symbols and regression line, and *S. squarrosum*, red symbols and regression line), one week after rewetting. CO_2 fluxes are net ecosystem exchange (NEE). With negative flux indicating sink of CO_2 from the air and positive fluxes release.

This scatter plot with regression lines shows that longer droughts lead to higher post-rewetting CO release (higher NEE) in both Sphagnum species, with S. palustre responding more strongly than S. squarrosum.

Murray, K. J., Tenhunen, J. D., & Nowak, R. S. (1993). Photoinhibition as a control on photosynthesis and production of Sphagnum mosses. *Oecologia*, 96(2), 200–207. https://doi.org/10.1007/BF00317733

Summary: This study examined photoinhibition in Sphagnum mosses under varying light and moisture conditions in a subalpine wetland, measuring chlorophyll fluorescence and gas exchange. The researchers found that photoinhibition significantly reduced photosynthetic capacity during periods of high light exposure, particularly when mosses were desiccated, with recovery occurring upon rehydration. The work was significant in demonstrating that light stress, rather than just water availability, can limit Sphagnum productivity in natural settings. A strength of the study is its field-based approach under natural conditions, though the relatively short observation period may not capture seasonal variation in photo inhibitory responses.

Relevance to my work: This research helps explain how light stress and water availability each limit Sphagnum photosynthesis, which is important for understanding changes in productivity under different environmental conditions.

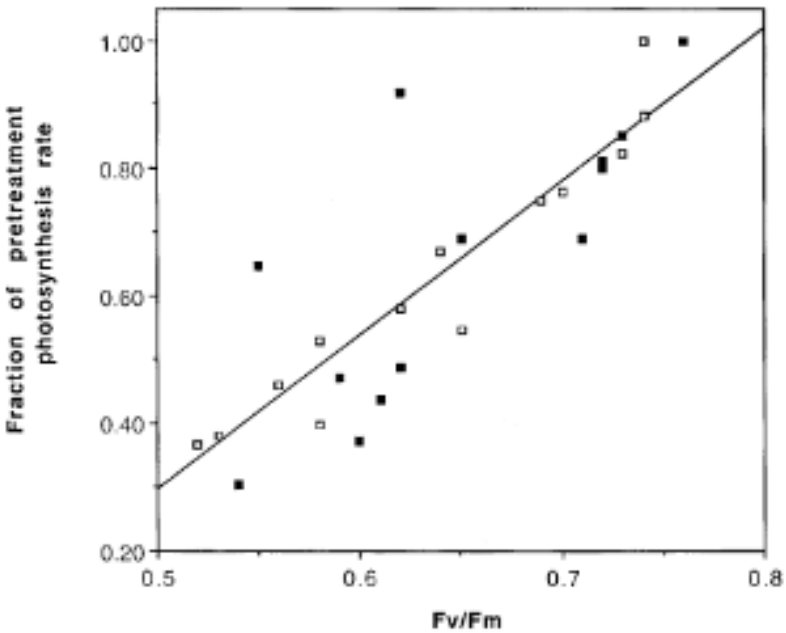


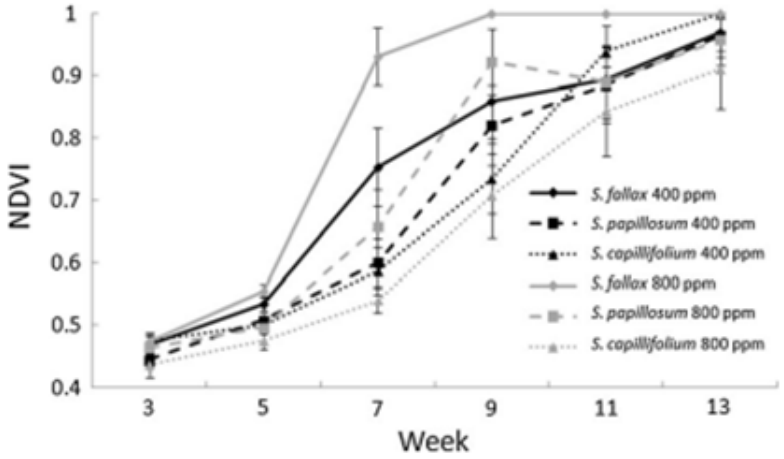
Fig. 5. The relationship between reduction in *Sphagnum* moss photosynthetic capacity and the ratio of variable to maximum chlorophyll fluorescence (F_v/F_m). Data obtained from growth chamber light treatments employing three irradiance levels: 150, 350, and 800 µmol photons m^{-2} s^{-1}. *Closed symbols:* arctic samples; *open symbols:* temperate samples

This is a scatter plot with a positive linear trend that shows photosynthetic efficiency (Fv/Fm) increases, Sphagnum photosynthetic capacity also increases. This means healthier moss (with higher Fv/Fm) can photosynthesize more, while light stress lowers both values.

Newman, T. R., Wright, N., Wright, B., & Sjögersten, S. (2018). Interacting effects of elevated atmospheric CO2 and hydrology on the growth and carbon sequestration of Sphagnum moss. *Wetlands Ecology and Management*, 26(5), 763–774. https://doi.org/10.1007/s11273-018-9607-x

Summary: Using controlled mesocosm experiments, the authors manipulated both CO concentrations (ambient vs. elevated) and water table depth to assess interactive effects on Sphagnum growth and carbon accumulation. They found that elevated CO stimulated Sphagnum growth only under optimal hydrological conditions, with the positive CO effect diminishing or disappearing when water tables were lowered, suggesting that hydrological stress may limit the capacity of peatlands to respond to rising atmospheric CO. The factorial design is a particular strength, though the relatively short duration (one growing season) may not capture longer-term acclimation or community-level feedbacks.

Relevance to my work: This study demonstrates the importance of considering multiple environmental factors simultaneously when assessing Sphagnum responses, highlighting that water availability may override other environmental drivers of productivity.



This is a line graph, it shows elevated CO (800 ppm) increases Sphagnum moss greenness and likely growth (as measured by NDVI) compared to current atmospheric levels (400 ppm), with the strongest positive response seen in S. fallax and more modest benefits in S. papillosum and S. capillifolium.

Norby, R. J., Baxter, T., Živković, T., & Weston, D. J. (2023). Shading contributes to Sphagnum decline in response to warming. *Ecology and Evolution*, 13(9), e10542-n/a. https://doi.org/10.1002/ece3.10542

Summary: The authors used open-top chambers in a bog to experimentally warm Sphagnum and monitored responses over multiple years, finding that warming led to increased vascular plant growth that shaded Sphagnum, reducing its photosynthesis and growth more than direct temperature effects. This work is significant in identifying indirect effects of warming mediated through competition for light rather than direct physiological temperature effects, highlighting the importance of community-level responses. A strength is the multi-year field experiment, though separating direct and indirect effects definitively would require additional manipulations of shading independent of warming.

Relevance to my work: This shows that shading from other plants can be just as important as temperature or moisture, highlighting the need to consider nearby vegetation when studying Sphagnum productivity.

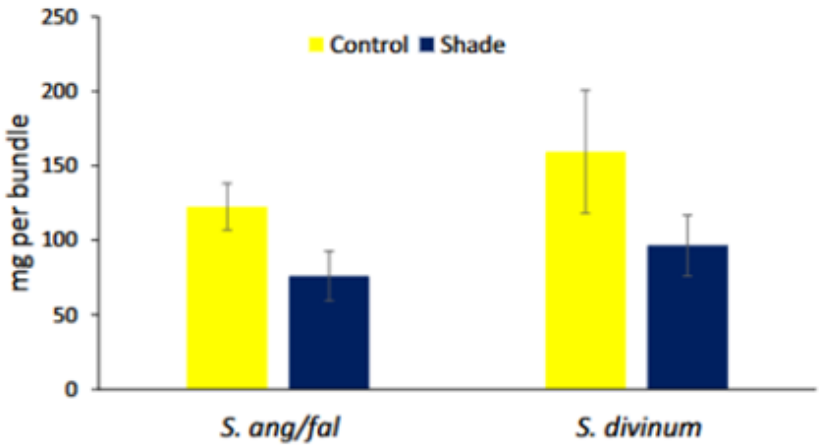


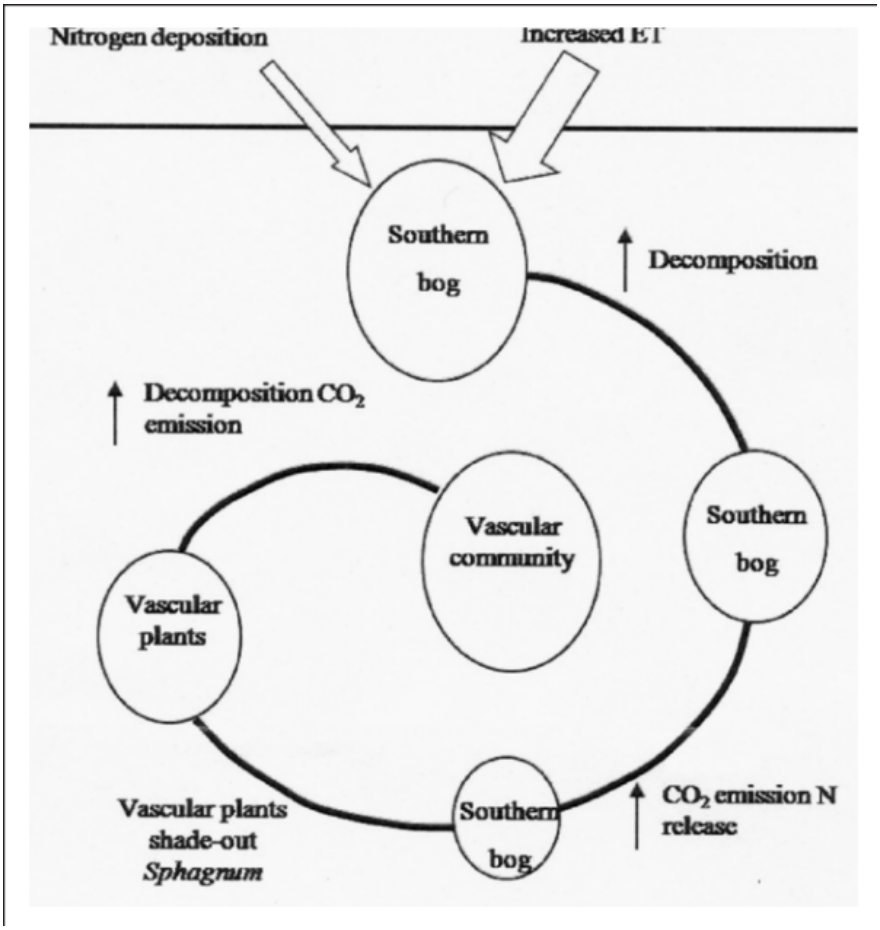
FIGURE 6 Dry weight increment of 10 *Sphagnum* stems in bundles (Figure 2c) under small cover of 90% shade cloth (Figure 2d) or control bundles with no shade cloth; shrubs were not removed. Data are the means over hollow and hummock locations in three blocks. Statistical significant for *S. angustifolia/fallax*: $p = .068$; *S. divinum*: $p = .202$.

This bar graph compares growth of two Sphagnum species in full light versus shade. Both species grew less when shaded, showing that light is important for moss growth, with the effect being stronger for *S. angustifolium/fallax*.

Schultheis, E. H., Hopfensperger, K. N., & Brenner, J. C. (2010). Potential Impacts of Climate Change on Sphagnum Bogs of the Southern Appalachian Mountains. *Natural Areas Journal*, 30(4), 417–424. https://doi.org/10.3375/043.030.0407

Summary: This study combined field surveys of Sphagnum distribution with climate modeling to project future habitat suitability for bog ecosystems in the Southern Appalachians. The authors found that increasing temperatures and altered precipitation patterns predicted by climate models could lead to substantial loss of suitable habitat for Sphagnum, particularly at lower elevations, threatening the persistence of these rare southern bog systems. The work is significant in highlighting regional vulnerability but is limited by uncertainty in downscaled climate projections and the assumption that species cannot adapt or migrate to maintain suitable microclimates.

Relevance to my work: This study shows how climate change may affect southern bogs, helping me understand how Sphagnum responds to stress in vulnerable locations.

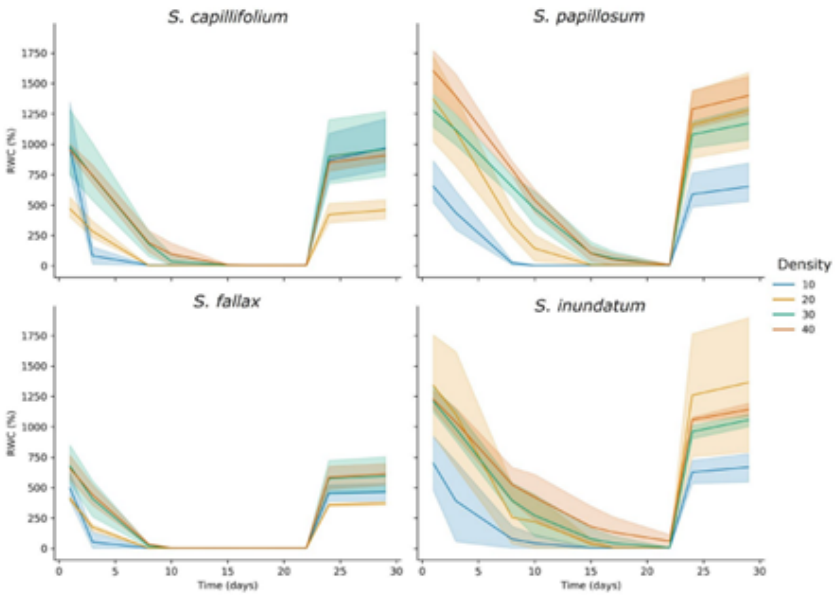


This flowchart diagram shows how climate change threatens southern Appalachian bogs through a cascade of effects. Diagram shows how increased evapotranspiration and nitrogen deposition lead to faster decomposition, more CO release, and conditions that favor vascular plants over Sphagnum moss.

Van de Koot, W. Q. M., Msonda, J., Olver, O. P., Doonan, J. H., & Nibau, C. (2024). Variation in Water-Holding Capacity in Sphagnum Species Depends on Both Plant and Colony Structure. *Plants (Basel)*, 13(8), 1061. https://doi.org/10.3390/plants13081061

Summary: This study measured water-holding capacity in 14 Sphagnum species by examining both individual plant traits (like capitulum structure and leaf shape) and how plants pack together in colonies, using morphological measurements, micro-CT imaging, and lab experiments. The researchers found that water-holding capacity varies significantly among species and depends on both plant structure and colony packing, with some species storing water internally while others rely on spaces between plants. This work is important because it explains how different structural features at multiple scales affect water retention in peatlands. A strength is the comprehensive approach examining both individual and colony-level traits, though lab conditions may not fully reflect field dynamics where factors like drying history also matter.

Relevance to my work: Understanding how plant structure affects water retention will guide my species selection and help explain differences in drought tolerance and productivity.



This multi-panel line graph tracks water content over a drought period for four Sphagnum species at different planting densities. The main finding is clear: denser plantings (more plants together) hold water longer during drought for all species. This shows that how tightly mosses are packed matters just as much as individual plant structure for surviving dry conditions.